Policy on Cyberwar Aerospace

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Executive Summary

Insights from Ukraine and Iran reveal how the U.S. should conduct cyber operations before and during wars.

Aerospace capabilities, critical to national security, would be secured through:

- Prioritizing network & supply chain resiliency over impenetrability.
- Operating offensively to gather intelligence & determine impacts of kinetic missions.
- Promoting cyber hygiene before wartime when adversaries prepare cyber strongholds.
- Increasing cyber vigilance when conflicts escalate.

Introduction

As a nonpartisan research organization, the Mitchell Institute for Aerospace Studies[1] emphasizes an aerospace perspective because operational effectiveness depends on secure space systems and communications. Policymakers must consider aerospace cyber resilience a national security priority.

Analysis of which operations were more vs less effective, their effects, their limits, and which defensive measures reduced their impact are provided for both Ukraine and Iran. These lessons help shape the best U.S. aerospace cyber policy recommended for protecting military readiness and national security across peacetime, run-up, and the war itself.

Case Study: Ukraine

Russian actors pre-positioned as early as March 2021 (before military invasion in February 2022) and sought "persistent access for strategic and battlefield intelligence" to facilitate destructive attacks across military, supply-chain vendors, and IT-providers [1]. It then became more visible and disruptive: DDoS[2] attacks were launched by DEV-0586[3] on Ukrainian government websites (01/14) and by GRU[4] on Ukrainian financial institutions (02/15), and IRIDIUM[5] deployed FoxBlade[6] (02/23). After war began, Russia expanded into a destructive wartime effort involving wipers, phishing, vulnerability exploitation, IT-provider compromises and espionage across Ukraine's networks [1, 2]. These operations were often tied to broader military activity, their greatest strategic value.

[1] https://www.mitchellaerospacepower.org/ – A nonpartisan research organization focused on fact-based policy analysis of aerospace power and U.S. national security.

[2] (Distributed-Denial-of-Service) A cyberattack where many systems overwhelm a server or network with traffic to make it unavailable to users.

[3] A Russian-linked cyber threat group identified by Microsoft that conducts Distributed-Denial-of-Service attacks against Ukrainian targets.

[4] (Glavnoye Razvedyvatelnoye Upravlenie) Russia's military intelligence agency responsible for cyber operations, espionage, and information warfare.

[5] A Russian-linked cyber threat group associated with military intelligence operations targeting Ukraine.

[6] Destructive malware deployed against Ukrainian organizations to delete data and disable systems.

Offensive Operations Employed

Figure 1. *Ranks 2021–2022 Russian offensive operations in Ukraine by effectiveness; legend provided below.*

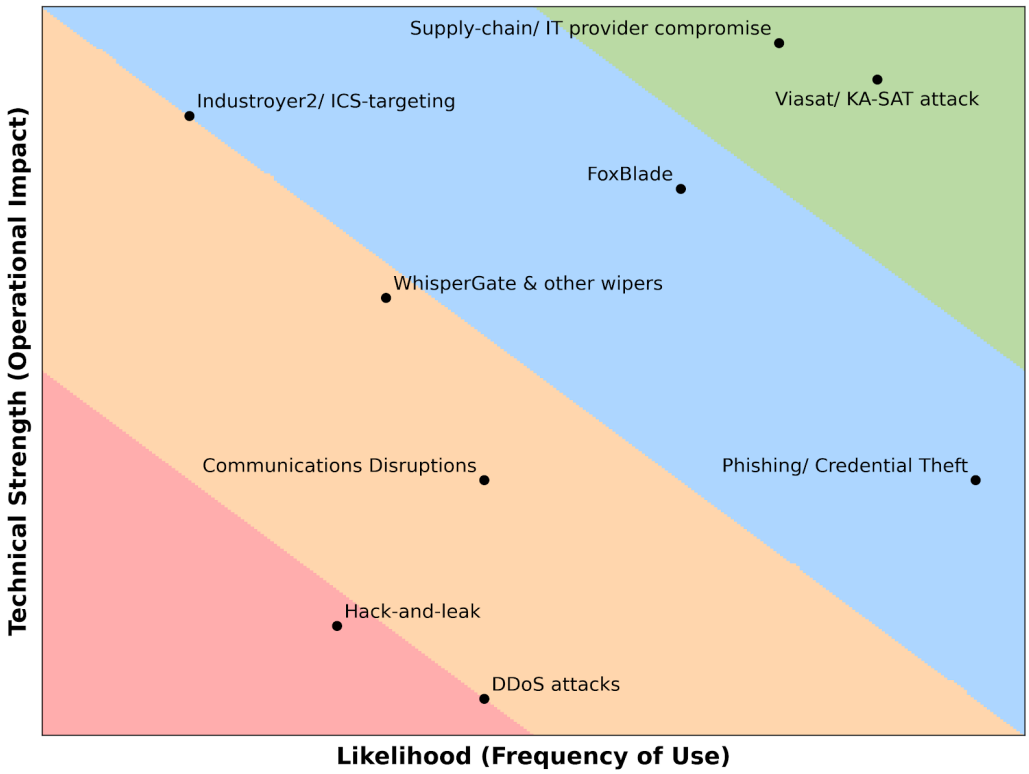
Operation	Target/Method	Effects	Effectiveness
Viasat / KA-SAT attack [3]	• Commercial satellite communications networks used by military & civilians • Exploited VPN misconfiguration & overwrote modem data	• "Partial interruption" of service, disrupting Ukrainian users & front-line communications [2, 4]	4
Supply-chain / IT-provider compromise [1, 5]	• Supply-chain vendors, IT providers, Ukrainian upstream services	• Enabled persistent access → destructive attacks & espionage against targets	4
FoxBlade [1]	• Ukrainian government, IT, energy & finance systems • Enterprise systems, not broad indiscriminate spread	• Damaged/disabled systems pre-invasion • Initial disruption • Limited real-world impact	3
Phishing / Credential Theft [1]	• Ukrainian government, military, media & NATO-country users [6] • Spoofed pages, phishing emails & credential collection	• Gained network access, allowed data collection • Supported later disruption	3
WhisperGate & other wipers [1]	• Ukraine government & IT systems • Broader networks over time using malware families	• Significant initial disruption • Less impact than prior Russian cyberattacks [2]	2
Communications Disruptions [4]	• Ukrainian telecom & ISP[7] • Disruptions against communications infrastructure	• Significant service outages: connectivity loss • Military effects limited/unclear	2
Industroyer2/ ICS[8]-targeting [1]	• Ukrainian energy infrastructure & utility networks	• Escalation toward physical disruption • Detected before major outages	2

[7] Internet Service Provider

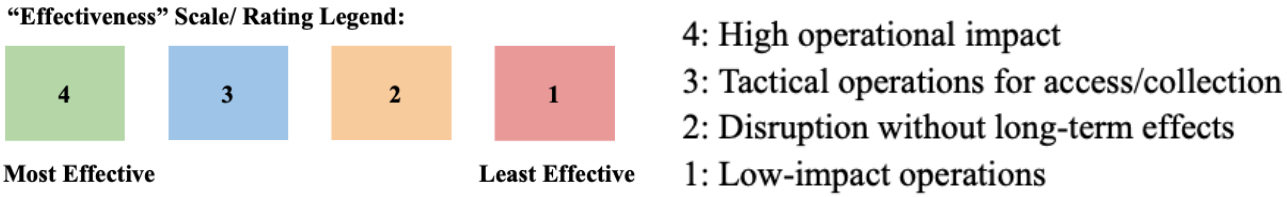
[8] (Industrial Control Systems): Systems controlling physical processes such as manufacturing or airfield operations.

	• Operational technology; not normal IT		
Hack-and-leak [2]	• Media organizations, public-facing platforms • Data leaks and exfiltration, deepfakes disinformation	• Shaped narratives and confusion • Minimal direct military impact	1
DDoS attacks [1]	• Ukrainian government websites & financial institutions • Traffic flooding to take services offline pre-invasion	• Disrupted access to online services • No decisive long-term effects	1

Figure 2[9]. *Visualizes offensive cyber operations based on Technical Strength*[10] *& Likelihood*[11].



[9] Note that Figure 1 and Figure 2 are based on the same analysis of the operations. The plotted points shown on Figure 2 are based on the "Effects" column in Figure 1 and were used in determining the rating of "Effectiveness".
[10] Technical Strength: Reflects operational impact of an attack like how much it directly degrades systems, communications, or battlefield effectiveness.
[11] Likelihood: How frequently or easily these operations were executed during the period of conflict.



While Russia's pre-positioning and early malware were aggressive, their impact hit a ceiling sooner than expected due to Ukrainian resilience and strong defensive preparation. Having faced Russian cyber activity since 2014, Ukraine hardened networks and built the "cyber tempo" needed for rapid recovery [7]. Support from private-sector tech giants provided advanced threat intelligence and backups, helping neutralize attacks like FoxBlade and WhisperGate before systemic failure [7, 8]. Russia found while they could break things, they couldn't keep them broken long enough to change the course of the war.

The Viasat attack highlights critical vulnerabilities in satellite communications and network architecture [3, 4]. It showed disabling a satellite network doesn't require kinetic weapons but can be achieved through vulnerabilities in the ground segment. An important lesson for aerospace policy: even sophisticated orbital assets are only as secure as the ground-based networks and routers managing them. Military communications rely on commercial providers, making them primary targets during peer-to-peer conflicts [6]. Ukraine's ability to switch systems proves aerospace resilience depends on redundant communication pathways, not any singular system [9].

U.S. policy needs to move past "impenetrable" defense and focus on mission continuity. Ukraine shows cyber attacks are now a "pre-phase" of warfare used to degrade air defenses and control-and-command (C2)[12] systems before conflict [3, 7]. This means securing aerospace supply chains and vendors as critical operational assets, preventing operational disruption [5].

Case Study: Iran

The 2025-2026 Iran conflict illustrates cyber operations becoming a critical enabler of aerospace warfare rather than a standalone tool. Cyber activity began before kinetic engagement, with Iranian actors conducting long-term reconnaissance, credential theft, and network pre-positioning, while U.S.-aligned operations focused on degrading Iranian air defenses, and sensor systems prior to strikes [10, 11]. During conflict, Iranian cyber efforts emphasized high-volume, low-sophistication attacks alongside targeted intelligence collection, including attempts to access surveillance systems and defense-related networks to support missile targeting and situational awareness [12, 13].

The most important lesson for U.S. aerospace policy is necessity of preemptive cyber shaping: disrupting adversary Integrated Air Defense Systems (IADS)[13], radar, and communications before kinetic strikes can

[12] (Command and Control) Systems that enable military leadership to direct forces and operations.

[13] (Integrated Air Defense System) A network of radar, missiles, and command systems used to detect and respond to aerial threats.

improve air superiority outcomes. Cyber must be integrated into operational planning as an early-phase force multiplier rather than a reactive tool.

The conflict exposed the vulnerability of defense industrial bases, supply chains, and Intelligence, Surveillance, and Reconnaissance (ISR)[14] ecosystem. Iranian actors targeted defense contractors and indirect vendors, exploiting weaker cybersecurity controls to gain access to sensitive systems [13]. The distributed nature of aerospace production creates systemic risk; subcontractor compromises provide pathways into high-value design and logistics platforms. Iranian-linked actors also exploited networked cameras and surveillance systems to collect real-time intelligence and assess strike effectiveness [12]. Aerospace operations depend heavily on interconnected ISR assets: radar, satellites, and airborne systems. If compromised, adversaries gain both tactical and strategic advantages without destroying physical assets.

Cyber operations proved most effective when synchronized with physical operations and least dangerous with resilient defenders. Iranian activity often coincided with missile and information campaigns, combining digital intrusion with psychological and kinetic effects [14], reinforcing the need for U.S. aerospace systems to assume simultaneous cyber and physical attack and build around mission continuity.

However, most incidents were limited in impact due to their relatively low sophistication [14], even though minor disruptions could degrade operations if systems lack redundancy. Identity-based vulnerabilities (credential theft and phishing), dominated Iranian operations over advanced exploits [12]. For U.S. aerospace policy, prioritizing multi-factor authentication, Zero Trust[15], continuous monitoring, rapid detection, segmentation, and recovery is imperative for ISR and mission effectiveness.

Offensive Operations Employed

Figure 3. *Ranks 2025–2026 Iranian offensive operations by effectiveness; legend based on Figure 2.*

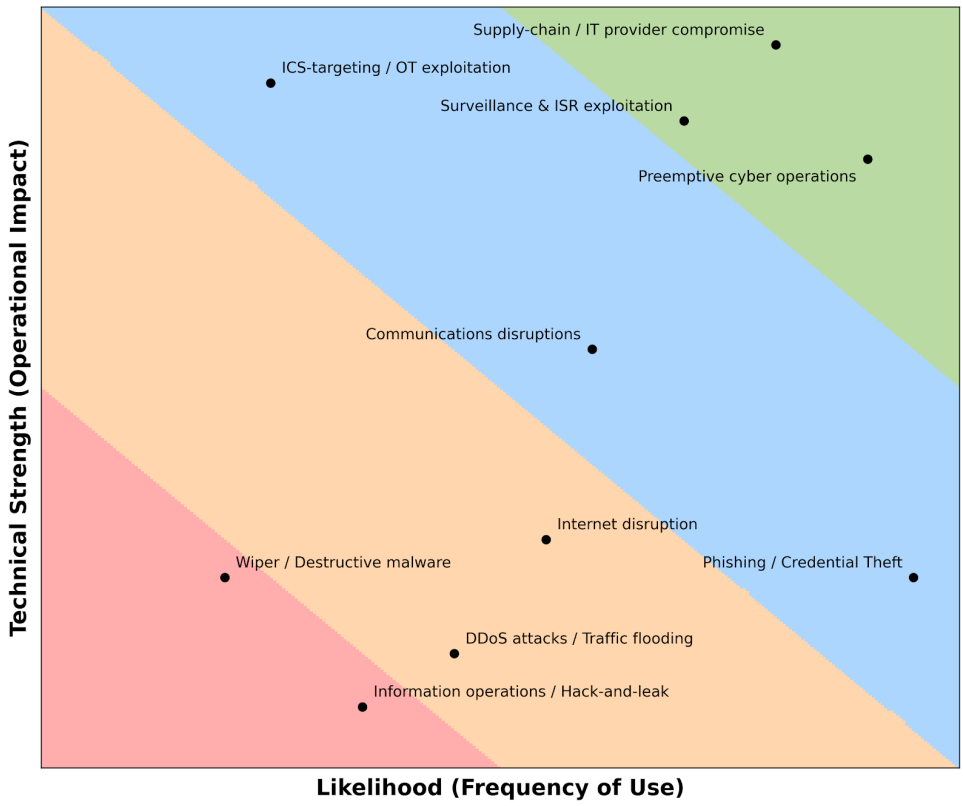
Operation	Target/Method	Observed Effect	Effectiveness
Supply-chain / IT-provider compromise	• Defense vendors, subcontractors [15]	• Long-term access into sensitive environments	4

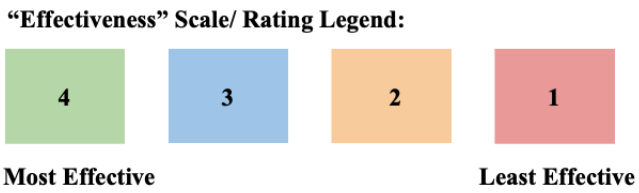
[14] (Intelligence, Surveillance, and Reconnaissance) Systems that collect and analyze battlefield information, such as satellites, drones, and radar.
[15] A security model where no user or system is trusted by default, even if they are inside the network.

ICS-targeting / OT exploitation	• Critical infrastructure; PLC[16]/SCADA[17] targeting [16]	• Operational disruptions & physical impact risk	4
Surveillance & ISR exploitation	• Surveillance cameras, monitoring systems; unauthorized access [17]	• Enabled intelligence collection & targeting awareness	4
Preemptive cyber operations	• Iranian air defense, C2, ISR [10]	• Reduced effectiveness of Iranian defenses Actor[18]	4
Communications disruptions	• Telecom networks, satellite communications [18]	• Degraded communications & coordination	3
Phishing / Credential Theft	• Defense contractors, government personnel; spear phishing [19]	• Persistent access & intelligence collection	3
Internet disruption	• National internet infrastructure [20]	• Reduced visibility & increased control	2
DDoS attacks / Traffic flooding	• U.S. government websites, financial systems, public-facing infrastructure • Traffic flooding, botnet-driven campaigns [19]	• Temporary disruption of services • Limited long-term impact [11]	2
Wiper / Destructive malware	• Critical infrastructure, enterprise systems • Wiper malware usage [17]	• Localized outages, data loss • Not strategic-level impact	1
Information operations / Hack-and-leak	• Media platforms, public systems [17]	• Narrative shaping & psychological impact	1

[16] (Programmable Logic Controller) A small industrial computer used to control physical machines and processes.
[17] (Supervisory Control and Data Acquisition) A high-level system that monitors and controls multiple industrial processes and controllers.
[18] An individual, group, or organization that performs a cyber activity or attack.

Figure 4. *Visualizes offensive cyber operations based on Technical Strength & Likelihood. Created based on guidelines in Figure 2.*





4: High operational impact
3: Tactical operations for access/collection
2: Disruption without long-term effects
1: Low-impact operations

Cyber capabilities were most valuable for intelligence collection and battlefield support, not large-scale disruption. For U.S. aerospace, this implies focusing on integrating cyber into ISR, targeting, and command systems while strengthening defensive resilience across networks.

Aerospace Special Treatment

The U.S. Air Force is critical in protecting American interests in peacetime and wartime. Understanding aerospace value is key to informing cyberwar policy. For 25 years, the Air Force was unchallenged in C2, allowing air dominance that ensures national security and facilitates military action globally [21]. Threats to this dominance jeopardize U.S. military response and remove the deterrence aerospace strength provides.

However, the cyber domain is vulnerable. Advanced persistent threats (APT)[19] exploit vulnerabilities in U.S. cyber systems and wait for strategic opportunities to attack [22]. Many Air Force systems rely on technology designed before cyber threats developed [23], making them more vulnerable. Other risks include attacks on infrastructure that support air base fueling, electricity, and secure communications [22]. Intelligence is highly desired by adversaries and cyber-attacks often target it. Supply chain attacks like Solar Winds[20] threaten C2 and reduce mission capabilities.

The Air Force addresses the threats through integrated cyber defenses focused on system resiliency [24], and Zero Trust strategies to manage systems [22]. Efforts like Cyber Squadron Initiative[21] have created teams to help counter cyber threats, but their effectiveness depends on continued expansion [25]. Transitioning to cyber-minded teams is essential to sustaining long-term U.S. air dominance.

U.S. Policy Recommendations

Recommendation by Phase

Peacetime	Staging	Wartime
Prevent adversarial prepositioning. ● Zero-trust implementation ● Supply chain security ● Continuous vulnerability management ● Establish & maintain "hunting" access in adversary air defense networks	Harden systems before conflict. ● Increase monitoring & alerting ● Integrate threat intelligence ● Network isolation & segmentation ● Early-phase disruptions to degrade adversary communications	Dedicate manpower to mission continuity. ● Rapid incident response execution ● Sustaining aerospace operations under attack ● Integrate cyberspace with ongoing operations ● Launch coordinated offensives to immediately disable adversary air defenses & C2

Peacetime

Peacetime cyber operations are imperative for sustained national security and effective cyber strikes during conflict. Cyber competition is driven by technological capability and intelligence collection, with both criminal and state-sponsored actors trying to find vulnerabilities to exploit.

In aerospace, threats primarily include ransomware, data theft, and intelligence operations to acquire new technologies. Defense-in-depth is essential, with multi-factor authentication, least privilege, and Zero Trust models reducing unauthorized access and limiting persistence. The Cybersecurity Maturity Model Certification ensures supply chain security for Controlled Unclassified Information.

[19] (Advanced Persistent Threat) A nation-state actor with access to extensive resources, funds, and manpower.
[20] A Russian-affiliated cyberattack that inserted malware into code run on various U.S. agency systems.
[21] The organizational shift of U.S. Air Force communications teams into dedicated cyber operations.

Adversaries now use AI to accelerate exploit discovery and attacks to machine speed. Defensive strategies must evolve while not leaving existing strategies behind. Continued public-private cooperation in sharing threat intelligence and vulnerability management will bridge the AI gap while organizations slowly adapt to the new playing field.

Staging

During staging, peacetime measures are insufficient against state-backed adversary threats. Interconnectivity between the aerospace industry and government allows organizations to leverage government intelligence channels to proactively brace for conflict.

Organizations must expand threat detection to neutralize destructive payloads before they reach their targets. Incident response plans become essential. System redundancy, cloud backups, manual fallbacks, and trained incident response teams allow for rapid recovery and minimized disruption.

Wartime

Alongside strengthening cyber systems against foreign attacks, aerospace stands apart in the private sector because it develops technology closely tied to armed forces and critical infrastructure. These connections lead to being important strategic targets, emphasizing the need for rapid incident response strategies to ensure aerospace operations can be sustained, even under concentrated attacks. Pre-wartime preparations is what separates the secure from the vulnerable.

Beyond defense, aerospace is key in supporting coordinated military offensives. Recent conflicts show the effectiveness of cyber operations in military mobilizations and battlefield support. The Russo-Ukrainian War has showcased how combined cyber-kinetic operations can be utilized to devastating results, and Iran's conflict shows how identity-based attacks and intelligence control dominate. With adversaries continuing to gain advantage in this domain, failing to adopt these tools would weaken national security.

Conclusion

U.S. policy must prioritize mission continuity, ensuring aerospace forces can fight through active technical assaults. This requires building redundant systems and securing the entire supply chain as a critical operational asset. The U.S. must institutionalize preemptive cyber operations to degrade adversary air defenses before kinetic engagement, ensuring aerospace dominance remains a guarantee, not an assumption.

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Appendix

Honor Code Statement: *We have complied with the generative AI policy of this course. Below is an record of all LLM usage from all group members.*

Prompt:
okay this is currently 480 words: The 2025–2026 Iran conflict illustrates how cyber operations have become a critical enabler of aerospace warfare rather than a standalone tool. Cyber activity began well before kinetic engagement, with Iranian actors conducting long-term reconnaissance, credential theft, and network pre-positioning, while U.S.-aligned operations focused on degrading Iranian air defenses, C2, and sensor systems prior to strikes (Citations 1,2). During the conflict, Iranian cyber efforts emphasized high-volume, low-sophistication attacks alongside targeted intelligence collection, including attempts to access surveillance systems and defense-related networks to support missile targeting and situational awareness (Citations 3,4).
The most important lesson for U.S. aerospace policy is the necessity of preemptive cyber shaping. Cyber operations targeting adversary integrated air defense systems (IADS) and ISR networks before kinetic strikes significantly improve air superiority outcomes. For example, reported disruptions to Iranian radar and communications infrastructure reduced their ability to respond effectively to aerial operations (C1). This demonstrates that cyber must be integrated into operational planning as an early-phase force multiplier rather than a reactive tool.
Another lesson is the vulnerability of the defense industrial base and supply chain. Iranian actors frequently targeted defense contractors and indirect vendors, exploiting weaker cybersecurity controls to gain access to sensitive systems (Citation 4). For aerospace, where production and maintenance rely on distributed suppliers, this creates systemic risk. A compromise in a subcontractor network can provide pathways into high-value systems such as aircraft design data or logistics platforms.
The conflict highlights the critical importance of protecting ISR and sensor ecosystems. Iranian-linked actors exploited networked cameras and surveillance systems to collect real-time intelligence and assess strike effectiveness (Citation 3). Aerospace operations depend heavily on interconnected ISR assets, including radar, satellites, and airborne systems. If these are compromised, adversaries gain both tactical and strategic advantages without needing to destroy physical assets.
Cyber operations proved most effective when synchronized with physical operations. Iranian cyber activity often coincided with missile and information campaigns, combining digital intrusion with psychological and kinetic effects (Citation 5). This reinforces the need for U.S. aerospace systems to operate under the assumption of simultaneous cyber and physical attack, requiring resilient architectures and mission continuity planning.
The conflict underscores that resilience outweighs prevention. Despite thousands of cyber incidents, most were limited in impact due to their relatively low sophistication (Citation 5). However, even minor disruptions can degrade operations if systems lack redundancy. Aerospace networks must prioritize rapid detection, segmentation, and recovery to sustain mission effectiveness under persistent attack.

Identity security emerged as a primary vulnerability. Many Iranian operations relied on credential theft and phishing rather than advanced exploits (Citation 3). This suggests that enforcing multi-factor authentication, Zero Trust architectures, and continuous monitoring could significantly reduce attack success rates.
Cyber capabilities were most valuable for intelligence collection and battlefield support, rather than large-scale disruption. For U.S. aerospace policy, this implies a strategic focus on integrating cyber into ISR, targeting, and command systems while strengthening defensive resilience across interconnected networks. --- can u cut it down to 400 or less please -- is there a better way to combine certain ideas/where is repetitveness here. With editing, do NOT change my tone, structure, style or any of the actual content that i have written down, i just need ot cut down REPTETITIVENESS ONLY. keep as much wording phases, and sentences from my original writing as much as possible except for slight changing/combining of certain ideas

Response:
Yes — there's definitely repetition you can compress.
The biggest overlap is:* **Sentence 1 + sentence 2**-- Both are making the same point that cyber is an early enabler/support tool rather than standalone.
* **Supply chain / ISR / synchronized ops / resilience / identity** – These are all good points, but right now each gets its own mini-paragraph, which makes the section feel a bit list-like.
* **Last sentence**-- It mostly restates what the earlier paragraphs already proved: cyber mattered most for intelligence collection, battlefield support, and resilience.
Best way to cut it
Keep the structure, but combine:
* intro + preemptive shaping
* resilience + identity security
* synchronized ops + battlefield support conclusion
That gets you under 400 without losing substance.

Revised version under 400 words
The 2025–2026 Iran conflict illustrates how cyber operations have become a critical enabler of aerospace warfare rather than a standalone tool. Cyber activity began well before kinetic engagement, with Iranian actors conducting long-term reconnaissance, credential theft, and network pre-positioning, while U.S.-aligned operations focused on degrading Iranian air defenses, C2, and sensor systems prior to strikes (Citations 1,2). During the conflict, Iranian cyber efforts emphasized high-volume, low-sophistication attacks alongside targeted intelligence collection, including attempts to access surveillance systems and defense-related networks to support missile targeting and situational awareness (Citations 3,4). The clearest lesson for U.S. aerospace policy is the importance of preemptive cyber shaping: disrupting adversary IADS, radar, and communications before kinetic strikes can improve air superiority outcomes and should be treated as an early-phase force multiplier rather than a reactive tool.

The conflict also exposed the vulnerability of the defense industrial base, supply chain, and ISR ecosystem. Iranian actors frequently targeted defense contractors and indirect vendors, exploiting weaker cybersecurity controls to gain access to sensitive systems (Citation 4). For

aerospace, where production and maintenance depend on distributed suppliers, this creates systemic risk, since a compromise in a subcontractor network can open pathways into aircraft design data, logistics platforms, or mission-support systems. At the same time, Iranian-linked actors exploited networked cameras and surveillance systems to collect real-time intelligence and assess strike effectiveness (Citation 3), showing how adversaries can gain tactical and strategic advantages by compromising radar, satellite-linked, and airborne ISR networks without destroying physical assets.

Iran also reinforces that cyber operations are most effective when synchronized with physical operations and least dangerous when defenders are resilient. Iranian activity often coincided with missile and information campaigns, combining digital intrusion with psychological and kinetic effects (Citation 5). Yet despite thousands of cyber incidents, most remained limited in impact because of their relatively low sophistication and the presence of redundancy, detection, segmentation, and recovery measures (Citation 5). Many operations still relied on credential theft and phishing rather than advanced exploits (Citation 3), which suggests that multi-factor authentication, Zero Trust architectures, and continuous monitoring could significantly reduce attack success rates. For U.S. aerospace policy, the broader lesson is clear: cyber is most valuable for intelligence collection and battlefield support, while resilience is what keeps attacks from becoming operationally decisive.

Word count

This version is **377 words**.

If you want, I can also do an even tighter **330–350 word** version so you have more room elsewhere.